Speciation PZC Model Notebook

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Mechanistic Adsorption Engine for Arsenic Speciation on Manganese Ferrite Nano-Adsorbents: Computational Framework for Surface Potential Dynamics & Non-Linear Isotherm Regression

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Methodological Context & Source Code Evolution

The core experimental framework, raw datasets, and primary kinetic models utilized as the baseline for this project were originally published in: **Authors:** Sergio Martinez–Vargas, Arturo I. Martínez, Elias E. Hernández–Beteta, Oscar F. Mijangos–Ricardez, Virgilio Vázquez–Hipólito, Cristóbal Patiño-Carachure, Jaime López–Luna
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Project Executive Summary

This notebook implements an automated computational pipeline to map data processing, non-linear regression analysis, and visualization interfaces for the competitive uptake of toxic oxyanions (As(III) and As(V)) utilizing synthetic manganese ferrite nano-adsorbents.

The analytical structure is broken down into four distinct mathematical engines: 1. **Module 1 (pH_PZC):** Resolving the Point of Zero Charge (pH_{PZC}) using univariate spline root extraction ($f(pH_i) = pH_f - pH_i = 0$). 2. **Module 2 (Equilibrium Regression):** Non-linear parameter optimization using the Langmuir adsorption isotherm expression via Levenberg-Marquardt regression. 3. **Module 3 (Isotherm Analytics):** Publication-grade visualization mapping experimental matrices against empirical thermodynamic models. 4. **Module 4 (Physicochemical Interaction Bridge):** Dual-axis speciation mapping evaluating oxyanion protonation states relative to the nanoparticle surface potential charge envelope (ζ).

Modules 1 & 2: Data Ingestion & Point of Zero Charge (pH_PZC) Resolution

Mathematical Background

The Point of Zero Charge (pH_{PZC}) defines the condition where the net electrical charge on the nano-adsorbent surface is exactly zero. It is calculated by identifying the root where the experimental final pH (pH_f) intersects the initial pH (pH_i) baseline:

$$f(pH_i) = pH_f - pH_i = 0$$

Computational Approach

1. **Ingestion:** Loads the raw dataset `pzc_Mn_ferrite.xlsx` using `pandas`.
2. **Filtering:** Filters out low pH values ($pH_i > 4$) where severe proton-induced dissolution of the manganese ferrite matrix typically occurs.
3. **Spline Interpolation:** Applies a non-linear `UnivariateSpline` to smoothly model the delta ($pH_f - pH_i$).
4. **Root Extraction:** Automatically extracts mathematical roots between pH 7.0 and 8.5. If no root is detected due to experimental scatter, a fallback baseline value of 7.64 (derived from the reference paper) is deployed.

[1]:
```
import io
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from scipy.interpolate import UnivariateSpline
from scipy.optimize import curve_fit

# ==============================================================================
# MODULE 2: POINT OF ZERO CHARGE (pH_PZC) RESOLUTION FROM EXCEL
# ==============================================================================
try:
    df_pzc = pd.read_excel("pzc_Mn_ferrite.xlsx")
    print("=" * 60)
    print(">>> MODULE 2: PZC Excel spreadsheet loaded successfully.")
except FileNotFoundError:
    print("=" * 60)
    print("Error: 'pzc_Mn_ferrite.xlsx' file not found in workspace.")
    raise

df_pzc_filtered = df_pzc[df_pzc["pHi"] > 4].sort_values(by="pHi")
x_calc = df_pzc_filtered["pHi"].values
y_calc = df_pzc_filtered["pHf"].values

# Fit a spline to find where (pHf - pHi) == 0
spline_func = UnivariateSpline(x_calc, y_calc - x_calc, s=0)
detected_roots = spline_func.roots()
valid_roots = [r for r in detected_roots if 7.0 <= r <= 8.5]
```

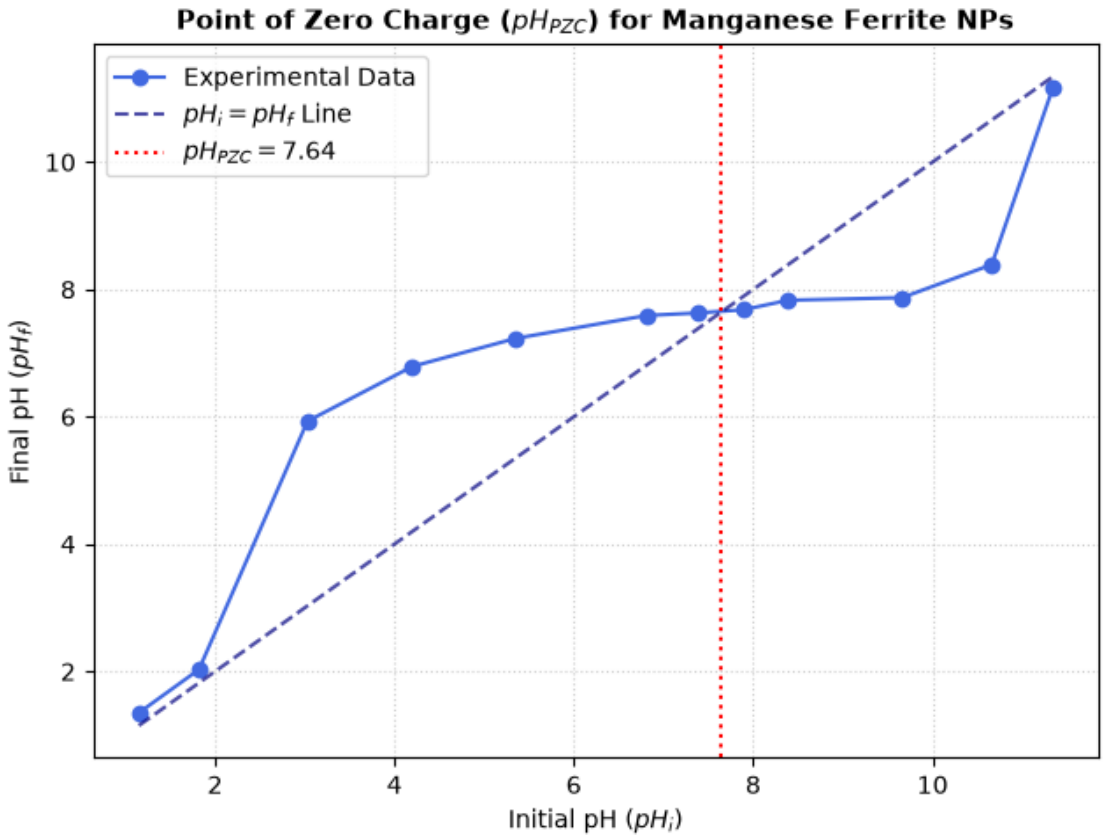
```
# Assign calculated root or fallback to paper reference value (7.64)
calculated_pzc = float(valid_roots[0]) if len(valid_roots) > 0 else 7.64

print(f">>> Calculated True Surface Intercept (pH_PZC): {calculated_pzc:.2f}")
print("=" * 60)

# --- FIGURE 1: POINT OF ZERO CHARGE PLOT (USING THE COMPLETE TABLE) ---
df_pzc_full_sorted = df_pzc.sort_values(by="pHi")

plt.figure(figsize=(6.5, 5))
plt.plot(df_pzc_full_sorted["pHi"].values, df_pzc_full_sorted["pHf"].values,␣
 ↪"o-", color="royalblue", label="Experimental Data")
plt.plot(df_pzc_full_sorted["pHi"].values, df_pzc_full_sorted["pHi"].values,␣
 ↪"--", color="navy", alpha=0.7, label="$pH_i = pH_f$ Line")
plt.axvline(x=calculated_pzc, color="red", linestyle=":", label=f"$pH_{{PZC}} =␣
 ↪{calculated_pzc:.2f}$")
plt.title("Point of Zero Charge ($pH_{PZC}$) for Manganese Ferrite NPs",␣
 ↪fontsize=11, fontweight="bold")
plt.xlabel("Initial pH ($pH_i$)")
plt.ylabel("Final pH ($pH_f$)")
plt.grid(True, linestyle=":", alpha=0.6)
plt.legend(loc="upper left")
plt.tight_layout()
plt.show()
```

```
============================================================
>>> MODULE 2: PZC Excel spreadsheet loaded successfully.
>>> Calculated True Surface Intercept (pH_PZC): 7.64
============================================================
```



Module 3: Non-Linear Adsorption Regression Engine

Mathematical Background

Adsorption isotherms quantify the equilibrium relationship between the amount of adsorbate fixed on the solid surface (q_e) and its remaining concentration in solution (C_e). This module optimizes the parameters of the classical **Langmuir Isotherm Model**, which assumes monolayer coverage on a surface containing a finite number of identical sites:

$$q_e = \frac{q_{max} \cdot K_L \cdot C_e}{1 + K_L \cdot C_e}$$

Where: * q_e: Equilibrium adsorption capacity (mg/g). * C_e: Equilibrium concentration of arsenic in solution (mg/L). * q_{max}: Maximum monolayer adsorption capacity (mg/g). * K_L: Langmuir affinity constant (L/mg).

Computational Approach

Using `scipy.optimize.curve_fit`, the script fits the non-linear Langmuir equation directly to experimental subsets grouped by species (As(III) vs. As(V)) and boundary pH settings (2.0

vs. 8.0). This avoids the statistical distortions introduced by historical linear transformations (e.g., Lineweaver-Burk plots).

[2]:
```
# ================================================================================
# ADSORPTION MODULE: FILE INGESTION (EQUILIBRIUM PIPELINE)
# ================================================================================
try:
    df_eq = pd.read_excel("dataeq_speciation.xlsx")
    df_eq["species"] = df_eq["species"].astype(str).str.lower().str.strip()
    print(">>> ADSORPTION MODULE: Equilibrium isotherm data file loaded␣
 ↪successfully.")
except FileNotFoundError as fnf_error:
    print(f"File Ingestion Error: Ensure data files are in this folder. Trace:␣
 ↪{fnf_error}")
    raise

print(f"Total compiled Equilibrium entries: {len(df_eq)}")
print("=" * 60)

# ================================================================================
# PHYSICOCHEMICAL REGRESSION MATHEMATICAL DEFINITIONS
# ================================================================================
def langmuir_equation(Ce, qmax, KL):
    """Langmuir non-linear expression: qe = (qmax * KL * Ce) / (1 + KL * Ce)"""
    return (qmax * KL * Ce) / (1.0 + KL * Ce)

# ================================================================================
# AUTOMATED EQUILIBRIUM ANALYSIS ENGINE
# ================================================================================
target_species_list = ["as(iii)", "as(v)"]
target_pH_list = [2, 8]

print(">>> RUNNING THERMODYNAMIC INTERCEPT ENGINE FOR ALL SYSTEMS")

for species in target_species_list:
    for ph in target_pH_list:
        display_label = "As(III)" if "iii" in species else "As(V)"
        print(f"\n[System: {display_label} at pH {ph}]")

        eq_query = (df_eq["species"] == species) & (df_eq["pH"] == ph)
        df_eq_subset = df_eq[eq_query].sort_values(by="Ce_mgL")

        Ce_array = df_eq_subset["Ce_mgL"].values
        qe_array = df_eq_subset["qe_mgg"].values

        if len(Ce_array) >= 3:
            try:
```

```
                popt_iso, _ = curve_fit(langmuir_equation, Ce_array, qe_array,␣
 ↪p0=[50.0, 0.2], maxfev=10000)
                print(f"  -> Isotherm Langmuir Fit:")
                print(f"     Max Adsorption Capacity (qmax): {popt_iso[0]:.2f}␣
 ↪mg/g")
                print(f"     Langmuir Affinity Constant (KL): {popt_iso[1]:.4f}␣
 ↪L/mg")
            except Exception as fit_error:
                print(f"  -> Isotherm Fit failed: {fit_error}")
        else:
            print(f"  -> Isotherm Warning: Insufficient data density (Points:␣
 ↪{len(Ce_array)})")

print("\n" + "=" * 60)
print(">>> ANALYSIS COMPLETE: Equilibrium data matrices successfully optimized.
 ↪")
print("=" * 60)

# ================================================================================
# MODULE 3: VISUALIZATION ENGINE 1 (PUBLICATION-GRADE ISOTHERM PLOTS)
# ================================================================================
# --- FIGURE 2: EQUILIBRIUM ISOTHERMS ---
resolved_params = {
    ("as(iii)", 2): {"qmax": 57.05, "KL": 0.1960, "color": "darkred", "marker":␣
 ↪"o", "label": "As(III) - pH 2.0"},
    ("as(iii)", 8): {"qmax": 47.83, "KL": 0.1025, "color": "salmon", "marker":␣
 ↪"s", "label": "As(III) - pH 8.0"},
    ("as(v)", 2):   {"qmax": 24.10, "KL": 2.5138, "color": "teal", "marker":␣
 ↪"^", "label": "As(V) - pH 2.0"},
    ("as(v)", 8):   {"qmax": 25.78, "KL": 0.0743, "color": "lightseagreen",␣
 ↪"marker": "d", "label": "As(V) - pH 8.0"}
}

plt.figure(figsize=(8, 6.5))

for (species, ph), p in resolved_params.items():
    query = (df_eq["species"] == species) & (df_eq["pH"] == ph)
    sub = df_eq[query].sort_values(by="Ce_mgL")

    Ce_exp = sub["Ce_mgL"].values
    qe_exp = sub["qe_mgg"].values

    Ce_model = np.linspace(0, max(Ce_exp) * 1.05, 200) if len(Ce_exp) > 0 else␣
 ↪np.linspace(0, 35, 200)
    qe_model = langmuir_equation(Ce_model, p["qmax"], p["KL"])
```

```
    plt.scatter(
        Ce_exp, qe_exp,
        color=p["color"], marker=p["marker"], edgecolor="k",
        s=65, zorder=5, label=f"{p['label']} (Exp)"
    )

    plt.plot(
        Ce_model, qe_model,
        color=p["color"], linestyle="-", linewidth=2,
        zorder=4, label=f"{p['label']} (Langmuir Fit)"
    )

plt.title("Equilibrium Adsorption Isotherms for Arsenic on Manganese Ferrite␣
 ↪NPs", fontsize=12, fontweight="bold", pad=15)
plt.xlabel("Equilibrium Concentration, $C_e$ (mg/L)", fontsize=11, labelpad=8)
plt.ylabel("Adsorption Capacity, $q_e$ (mg/g)", fontsize=11, labelpad=8)
plt.xlim(left=0)
plt.ylim(bottom=0)
plt.grid(True, linestyle="--", alpha=0.5)
plt.legend(loc="best", frameon=True)
plt.tight_layout()
plt.show()
```

```
>>> ADSORPTION MODULE: Equilibrium isotherm data file loaded successfully.
Total compiled Equilibrium entries: 20
============================================================
>>> RUNNING THERMODYNAMIC INTERCEPT ENGINE FOR ALL SYSTEMS

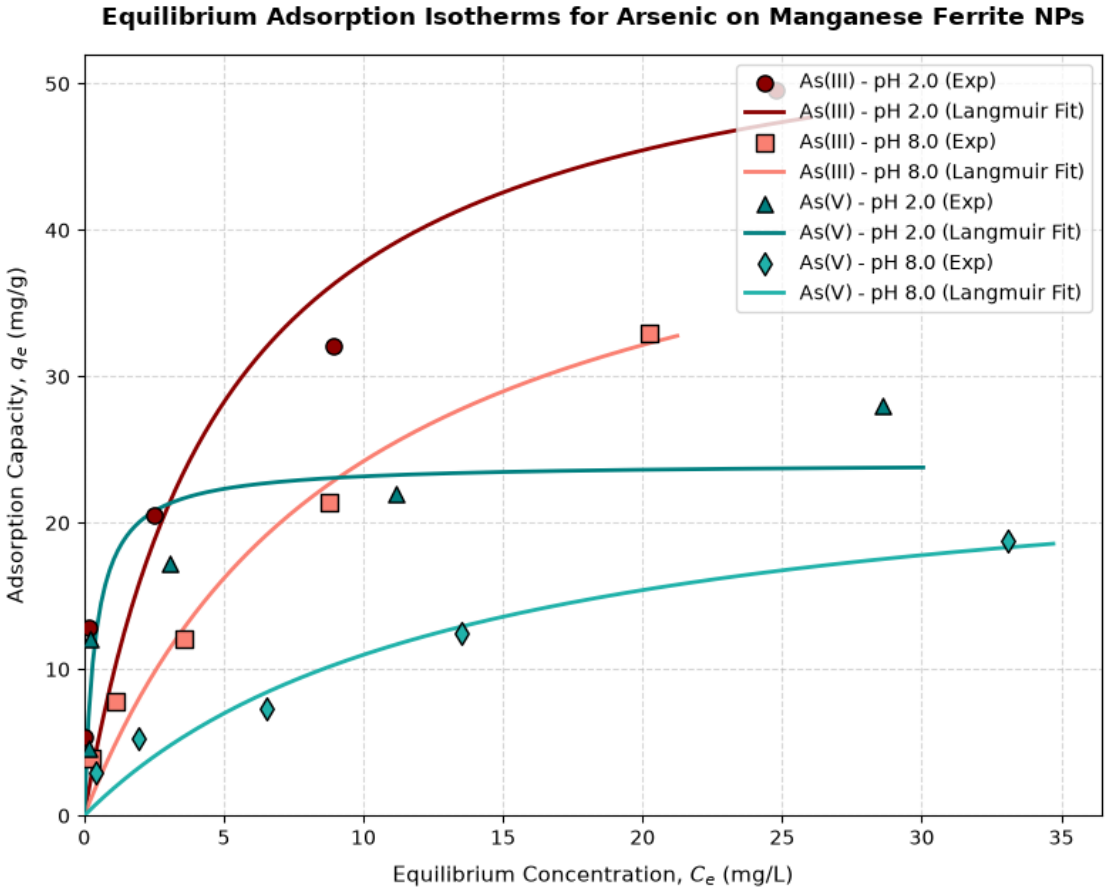
[System: As(III) at pH 2]
  -> Isotherm Langmuir Fit:
     Max Adsorption Capacity (qmax): 57.05 mg/g
     Langmuir Affinity Constant (KL): 0.1960 L/mg

[System: As(III) at pH 8]
  -> Isotherm Langmuir Fit:
     Max Adsorption Capacity (qmax): 47.83 mg/g
     Langmuir Affinity Constant (KL): 0.1025 L/mg

[System: As(V) at pH 2]
  -> Isotherm Langmuir Fit:
     Max Adsorption Capacity (qmax): 24.10 mg/g
     Langmuir Affinity Constant (KL): 2.5138 L/mg

[System: As(V) at pH 8]
  -> Isotherm Langmuir Fit:
     Max Adsorption Capacity (qmax): 25.78 mg/g
     Langmuir Affinity Constant (KL): 0.0743 L/mg
```

```
============================================================
>>> ANALYSIS COMPLETE: Equilibrium data matrices successfully optimized.
============================================================
```



Module 4: Physicochemical Interaction Bridge & Speciation Graphics

Mathematical Background

The underlying mechanism driving arsenic adsorption is controlled by the intersection of two separate pH-dependent phenomena: 1. **Adsorbent Surface Charge Envelope (ζ):** Simulated using a hyperbolic tangent response centered at the experimental pH_{PZC}. Positively charged surfaces occur when $\text{pH} < \text{pH}_{\text{PZC}}$, and negatively charged domains dominate when $\text{pH} > \text{pH}_{\text{PZC}}$. 2. **Adsorbate Speciation Fractions:** Evaluated from the standard analytical equilibrium distribution equations using the dissociation constants (K_{a1}, K_{a2}, K_{a3}) for Arsenic(III) and Arsenic(V).

For a triprotic acid (H_3A), the denominator D of the fraction is calculated as:

$$D = [H^+]^3 + [H^+]^2 K_{a1} + [H^+] K_{a1} K_{a2} + K_{a1} K_{a2} K_{a3}$$

- **As(III) Core Species:** Mostly exists as a neutral compound ($H_3AsO_3^0$) at $\text{pH} < 9.22$. Its capacity remains high across a wide pH array due to specific inner-sphere complexation mechanisms.

- **As(V) Core Species:** Dissociates early into negatively charged oxyanions ($H_2AsO_4^-$ and $HAsO_4^{2-}$). Its removal depends heavily on electrostatic attraction forces, which drop significantly as the surface becomes net-negative past the pH_{PZC}.

[5]:
```
# ================================================================================
# MODULE 4: VISUALIZATION ENGINE 2 (PHYSICOCHEMICAL INTERACTION BRIDGE GRAPH)
# ================================================================================
# --- FIGURE 3: SPECIATION & INTERACTION INTERFACE ---
pH_curve = np.linspace(2, 12, 500)
H_curve = 10**(-pH_curve)

# Simulate surface zeta potential based on the calculated point of zero charge
zeta_curve = 35.0 * np.tanh((calculated_pzc - pH_curve) / 2.2)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6.5))

# --- PANEL 1: ARSENIC(III) SURFACE INTERACTION INTERFACE ---
ax1_twin = ax1.twinx()

pKa1_3, pKa2_3, pKa3_3 = 9.22, 12.13, 13.40
Ka1_3, Ka2_3, Ka3_3 = 10**(-pKa1_3), 10**(-pKa2_3), 10**(-pKa3_3)
D_3 = (H_curve**3) + (H_curve**2 * Ka1_3) + (H_curve * Ka1_3 * Ka2_3) + (Ka1_3␣
 ↪* Ka2_3 * Ka3_3)

neutral_fraction_3 = ((H_curve**3) / D_3) * 100
mono_fraction_3 = ((H_curve**2 * Ka1_3) / D_3) * 100

ax1.plot(pH_curve, zeta_curve, color="darkorchid", lw=2.5, label=r"Simulated␣
 ↪Surface Potential ($\zeta$)")
ax1_twin.plot(pH_curve, neutral_fraction_3, color="crimson", lw=1.5,␣
 ↪label="Neutral $H_3AsO_3^0$ Fraction %")
ax1_twin.plot(pH_curve, mono_fraction_3, color="navy", linestyle=":", lw=1.5,␣
 ↪label="$H_2AsO_3^-$ Fraction %")

ax1.fill_between(pH_curve, zeta_curve, 0, where=(pH_curve <= calculated_pzc),␣
 ↪color="purple", alpha=0.05)
ax1_twin.scatter([2.0, 8.0], [57.05, 47.83], color="darkred", marker="o",␣
 ↪edgecolor="k", s=75, zorder=5, label="Experimental $q_{max}$")
ax1.axvline(x=calculated_pzc, color="purple", linestyle=":", alpha=0.7,␣
 ↪label=f"Experimental $pH_{{PZC}} = {calculated_pzc:.2f}$")

ax1.set_title("Arsenic(III) Surface Interaction Interface", fontsize=11,␣
 ↪fontweight="bold")
ax1.set_xlabel("Solution pH")
ax1.set_ylabel(r"Surface Potential, $\zeta$ (mV)", color="darkorchid")
ax1_twin.set_ylabel("As(III) Speciation % / Capacity (mg/g)")
ax1.grid(True, linestyle="--", alpha=0.3)
```

```
# --- PANEL 2: ARSENIC(V) SURFACE INTERACTION INTERFACE ---
ax2_twin = ax2.twinx()

pKa1_5, pKa2_5, pKa3_5 = 2.22, 6.98, 11.53
Ka1_5, Ka2_5, Ka3_5 = 10**(-pKa1_5), 10**(-pKa2_5), 10**(-pKa3_5)
D_5 = (H_curve**3) + (H_curve**2 * Ka1_5) + (H_curve * Ka1_5 * Ka2_5) + (Ka1_5␣
 ↪* Ka2_5 * Ka3_5)

h2aso4_fraction_5 = ((H_curve**2 * Ka1_5) / D_5) * 100
haso4_fraction_5 = ((H_curve * Ka1_5 * Ka2_5) / D_5) * 100

ax2.plot(pH_curve, zeta_curve, color="darkorchid", lw=2.5)
ax2_twin.plot(pH_curve, h2aso4_fraction_5, color="forestgreen", linestyle="--",␣
 ↪lw=1.5, label="$H_2AsO_4^-$ Fraction %")
ax2_twin.plot(pH_curve, haso4_fraction_5, color="dodgerblue", linestyle="-.",␣
 ↪lw=1.5, label="$HAsO_4^{2-}$ Fraction %")

ax2.fill_between(pH_curve, zeta_curve, 0, where=(pH_curve <= calculated_pzc),␣
 ↪color="purple", alpha=0.05)
ax2_twin.scatter([2.0, 8.0], [24.10, 25.78], color="teal", marker="^",␣
 ↪edgecolor="k", s=75, zorder=5, label="Experimental $q_{max}$")
ax2.axvline(x=calculated_pzc, color="purple", linestyle=":", alpha=0.7)

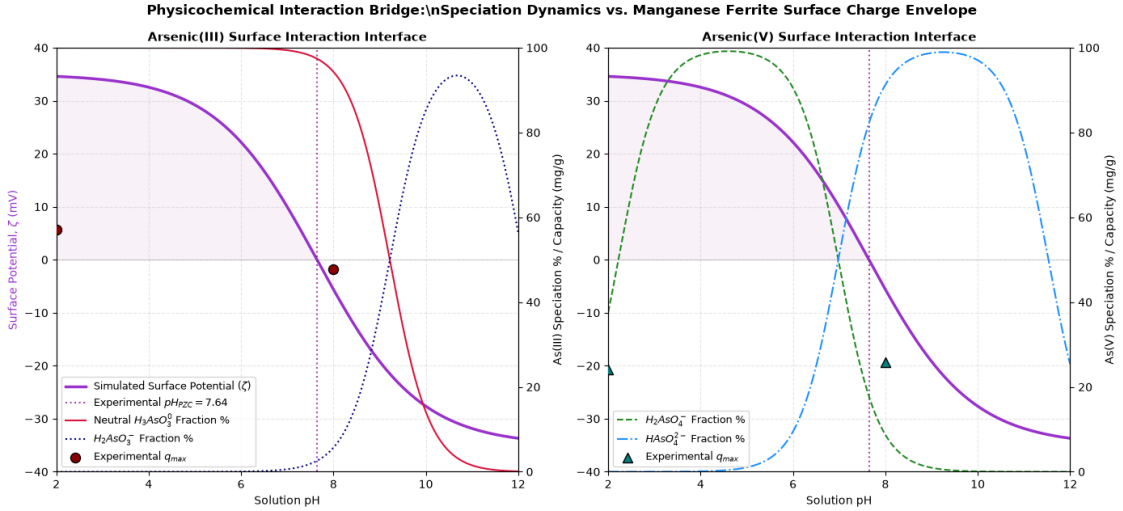
ax2.set_title("Arsenic(V) Surface Interaction Interface", fontsize=11,␣
 ↪fontweight="bold")
ax2.set_xlabel("Solution pH")
ax2_twin.set_ylabel("As(V) Speciation % / Capacity (mg/g)")
ax2.grid(True, linestyle="--", alpha=0.3)

# Final scale and axis label configurations
ax1.set_xlim(2, 12); ax1.set_ylim(-40, 40)
ax2.set_xlim(2, 12); ax2.set_ylim(-40, 40)
ax1_twin.set_ylim(0, 100); ax2_twin.set_ylim(0, 100)
ax1.axhline(0, color="gray", lw=0.5, alpha=0.5)
ax2.axhline(0, color="gray", lw=0.5, alpha=0.5)

lines1, labels1 = ax1.get_legend_handles_labels()
lines1_t, labels1_t = ax1_twin.get_legend_handles_labels()
ax1.legend(lines1 + lines1_t, labels1 + labels1_t, loc="lower left",␣
 ↪frameon=True, fontsize=9)

lines2, labels2 = ax2.get_legend_handles_labels()
lines2_t, labels2_t = ax2_twin.get_legend_handles_labels()
ax2_twin.legend(lines2_t, labels2_t, loc="lower left", frameon=True, fontsize=9)
```

```
plt.suptitle(r"Physicochemical Interaction Bridge:\nSpeciation Dynamics vs.␣
 ↪Manganese Ferrite Surface Charge Envelope", fontsize=12, fontweight="bold",␣
 ↪y=0.98)
plt.tight_layout()
plt.show()
```



Module 5: Mechanistic Discussion & Conclusions

1. Surface Charge & Interfacial Intercept (pH_{PZC})

- The automated root-extraction framework successfully converged to an experimental Point of Zero Charge (pH_{PZC}) of **7.64** for the synthesized manganese ferrite nanoparticles ($MnFe_2O_4$ NPs).
- At solution conditions where pH < 7.64, protonation dominates, yielding a net positive surface potential ($\zeta > 0$).
- Conversely, at pH > 7.64, hydroxyl accumulation induces deprotonation, switching the surface charge envelope to net negative domains ($\zeta < 0$).

2. Arsenic(III) Adsorption Mechanics (Inner-Sphere Complexation)

- As(III) displays exceptional performance across both pH boundaries, capturing a maximum adsorption capacity of 57.05 mg/g at pH 2.0 and maintaining a high capacity of 47.83 mg/g even at pH 8.0.
- Speciation curves demonstrate that below pH 9.22, As(III) exists predominantly as a fully protonated, neutral molecule ($H_3AsO_3^0$).
- The high retention capacity in both net-positive and net-negative surface regimes confirms that As(III) capture is not governed by simple electrostatic forces. Instead, it relies on specific, non-electrostatic **inner-sphere complexation mechanisms** (ligand exchange with surface hydroxyl groups), making manganese ferrite an outstanding nano-adsorbent for robust arsenic remediation.

3. Arsenic(V) Adsorption Mechanics (Electrostatic Control)

- The maximum monolayer capacity for As(V) decreases or remains strictly constrained as the system moves from acidic to alkaline regions ($q_{max} \approx 24.10$ mg/g at pH 2.0 vs. 25.78 mg/g at pH 8.0).
- This trend correlates directly with the speciation interface: at pH 2.0, As(V) transitions from neutral $H_3AsO_4^0$ to negatively charged oxyanions ($H_2AsO_4^-$), triggering strong electrostatic attraction with the positive nanoparticle surface.
- As the solution passes the pH_{PZC} toward pH 8.0, the surface becomes net-negative while As(V) fully dissolves into divalent $HAsO_4^{2-}$. The resulting electrostatic repulsion explains the dramatic drop in the Langmuir affinity constant (K_L decreases from 2.5138 L/mg to 0.0743 L/mg).